

TIME SERIES ANALYSIS & CLASSIFICATION

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Chapter VIII: Forecasting

1. Introduction
2. Deterministic Trend Models
3. *ARIMA* Models

INTRODUCTION

Introduction

- ▶ We have discussed two types of statistical models for time series data:
 - Deterministic trend models of the form $Y_t = \mu_t + X_t$, where $\{X_t\}$ is a zero-mean stochastic process.
 - $ARIMA(p, d, q)$ models of the form $\phi(B)(1 - B)^d Y_t = \theta(B)e_t$, where $\{e_t\}$ is a zero-mean white noise.
- ▶ For both types of models, we have studied:
 - Model specification
 - Model fitting
 - Diagnostic procedures to assess model fit

Introduction

Preview: We now switch our attention to forecasting.

- ▶ We start with a sample of process values up until time t , say, $Y_1, \dots, Y_t$. These are our observed data.
- ▶ Forecasting refers to predicting future values of the process, i.e., $Y_{t+1}, Y_{t+2}, Y_{t+3}, \dots$
- ▶ In general, Y_{t+l} is the value of the process at time $t + l$, where $l \geq 1$.
- ▶ We call t the forecast origin and l the lead time. Y_{t+l} is " l steps ahead" of the most recently observed value Y_t .

Important

- ▶ By "forecasting," we mean predicting the value of a future random variable Y_{t+l} .
- ▶ Prediction is a more challenging problem than estimating a population (model) parameter.
- ▶ Model parameters are fixed (but unknown) values. Random variables are not fixed; they are random.

Approach

- ▶ We need to adopt a formal mathematical criterion to calculate model forecasts. The criterion we will use is based on the **mean squared error of prediction**:

$$MSEP = E \left([Y_{t+l} - h(Y_1, \dots, Y_t)]^2 \right)$$

- ▶ Suppose we have a sample of observed data $Y_1, \dots, Y_t$ and want to predict Y_{t+l} . We choose the function $h(Y_1, \dots, Y_t)$ that minimizes $MSEP$. This function will be our forecasted value of Y_{t+l} .

Approach

- ▶ The general solution to this minimization problem is:

$$h(Y_1, \dots, Y_t) = E(Y_{t+l} | Y_1, \dots, Y_t)$$

- ▶ This is called the minimum mean squared error (*MMSE*) forecast:

$$\hat{Y}_t(l) = E(Y_{t+l} | Y_1, \dots, Y_t)$$

Conditional Expectation Rules

- ▶ The conditional expectation $E(Z|Y_1, \dots, Y_t)$ is a function of $Y_1, \dots, Y_t$.
- ▶ If c is a constant, then $E(c|Y_1, \dots, Y_t) = c$.
- ▶ If Z_1 and Z_2 are random variables, then

$$E(Z_1 + Z_2|Y_1, \dots, Y_t) = E(Z_1|Y_1, \dots, Y_t) + E(Z_2|Y_1, \dots, Y_t)$$

i.e., conditional expectation is additive (similar to unconditional expectation).

Conditional Expectation Rules

- ▶ If Z is a function of $Y_1, \dots, Y_t$, say, $Z = f(Y_1, \dots, Y_t)$, then

$$E(Z | Y_1, \dots, Y_t) = E[f(Y_1, \dots, Y_t) | Y_1, \dots, Y_t] = f(Y_1, \dots, Y_t)$$

In other words, once you condition on $Y_1, \dots, Y_t$, any function of them acts as a constant.

- ▶ If Z is independent of $Y_1, \dots, Y_t$, then

$$E(Z | Y_1, \dots, Y_t) = E(Z)$$

DETERMINISTIC TREND MODELS

Linear Trend Model

Consider the model $Y_t = \mu_t + X_t$, where:

- ▶ μ_t is a deterministic (non-random) trend function.
- ▶ $\{X_t\}$ is a white noise process with $E(X_t) = 0$ and $\text{var}(X_t) = \gamma_0$ (constant).

Example: Linear Trend Model

- By direct calculation, the l -step ahead forecast is:

$$\begin{aligned}\hat{Y}_t(l) &= E(Y_{t+l} \mid Y_1, \dots, Y_t) \\ &= E(\mu_{t+l} + X_{t+l} \mid Y_1, \dots, Y_t) \\ &= E(\mu_{t+l} \mid Y_1, \dots, Y_t) + E(X_{t+l} \mid Y_1, \dots, Y_t) \\ &= \mu_{t+l} + 0 \\ &= \mu_{t+l}\end{aligned}$$

because μ_{t+l} is non-random and X_{t+l} is a zero-mean random variable independent of $Y_1, \dots, Y_t$.

Example: Linear Trend Model

- Therefore, the *MMSE* forecast is :

$$\hat{Y}_t(l) = \mu_{t+l}$$

- For example, if $\mu_t = \beta_0 + \beta_1 t$, a linear trend model, then:

$$\begin{aligned}\hat{Y}_t(l) &= \mu_{t+l} = \beta_0 + \beta_1(t+l) \\ &= \beta_0 + \beta_1 t + \beta_1 l\end{aligned}$$

Remark

- ▶ One major drawback with predictions made from deterministic trend models is that they are based only on the least squares model fit.
- ▶ That is, the forecast for Y_{t+l} ignores the correlation between Y_{t+l} and $Y_1, \dots, Y_t$. Therefore, the analyst who makes these predictions is ignoring this correlation and, in addition, is assuming that the fitted trend is applicable indefinitely into the future; i.e., "the trend lasts forever."

Forecast Error in Deterministic Trend Models

Terminology: For deterministic trend models of the form

$$Y_t = \mu_t + X_t,$$

where $E(X_t) = 0$ and $\text{var}(X_t) = \gamma_0$ (constant), the forecast error at lead time l , denoted by $e_t(l)$, is:

$$e_t(l) = Y_{t+l} - \hat{Y}_t(l) = \mu_{t+l} + X_{t+l} - \mu_{t+l} = X_{t+l}.$$

► For all $l \geq 1$:

$$\begin{aligned} E[e_t(l)] &= E(X_{t+l}) = 0, \\ \text{var}[e_t(l)] &= \text{var}(X_{t+l}) = \gamma_0. \end{aligned}$$

Forecast Error in Deterministic Trend Models

- ▶ Forecasts are unbiased since the forecast error is an unbiased estimator of 0.
- ▶ The forecast error variance remains constant for all lead times l .
- ▶ These results aid in deriving prediction intervals for future values.

ARIMA MODELS

ARIMA Models

- We now discuss forecasting methods with *ARIMA* models. Recall that an *ARIMA*(p, d, q) process can be written generally as

$$\phi(B)(1 - B)^d Y_t = \theta_0 + \theta(B)e_t,$$

where θ_0 is an intercept term.

- We first focus on stationary *ARMA*(p, q) models, that is, *ARIMA*(p, d, q) models with $d = 0$. Special cases are treated in detail.

$AR(1)$ Model

- Consider the $AR(1)$ model:

$$Y_t - \mu = \phi(Y_{t-1} - \mu) + e_t$$

where the overall (process) mean $\mu = E(Y_t)$, and $\{e_t\}$ is zero mean white noise with $\text{Var}(e_t) = \sigma_e^2$.

1-step ahead Forecast

- The *MMSE* forecast of Y_{t+1} , the 1-step ahead forecast, is:

$$\begin{aligned}\hat{Y}_t(1) &= E(Y_{t+1} | Y_1, \dots, Y_t) = E[\mu + \phi(Y_t - \mu) + e_{t+1} | Y_1, \dots, Y_t] \\ &= E(\mu | Y_1, \dots, Y_t) + E[\phi(Y_t - \mu) | Y_1, \dots, Y_t] \\ &\quad + E(e_{t+1} | Y_1, \dots, Y_t)\end{aligned}$$

- From the properties of conditional expectation:

$$\begin{aligned}E(\mu | Y_1, \dots, Y_t) &= \mu \quad (\text{since } \mu \text{ is a constant}) \\ E[\phi(Y_t - \mu) | Y_1, \dots, Y_t] &= \phi(Y_t - \mu) \\ E(e_{t+1} | Y_1, \dots, Y_t) &= E(e_{t+1}) = 0,\end{aligned}$$

(since e_{t+1} is independent of $Y_1, \dots, Y_t$)

1-step ahead Forecast

- Therefore, the MMSE forecast of Y_{t+1} is:

$$\hat{Y}_t(1) = \mu + \phi(Y_t - \mu)$$

2-step ahead forecast

- ▶ The *MMSE* forecast of Y_{t+2} , the 2-step ahead forecast, is

$$\begin{aligned}\hat{Y}_t(2) &= E(Y_{t+2} | Y_1, \dots, Y_t) \\ &= E[\mu + \phi(Y_{t+1} - \mu) + e_{t+2} | Y_1, \dots, Y_t] \\ &= E(\mu | Y_1, \dots, Y_t) + E[\phi(Y_{t+1} - \mu) | Y_1, \dots, Y_t] \\ &\quad + E(e_{t+2} | Y_1, \dots, Y_t)\end{aligned}$$

- ▶ Now, we have

$$\begin{aligned}E[\phi(Y_{t+1} - \mu) | Y_1, \dots, Y_t] &= E\{\phi[\mu + \phi(Y_t - \mu) + e_{t+1} - \mu] | Y_1, \dots, Y_t\} \\ &= E\{\phi[\phi(Y_t - \mu) + e_{t+1}] | Y_1, \dots, Y_t\} \\ &= E[\phi^2(Y_t - \mu) | Y_1, \dots, Y_t] \\ &\quad + E(\phi e_{t+1} | Y_1, \dots, Y_t).\end{aligned}$$

2-step ahead forecast

- ▶ From the properties of conditional expectation, we again note the following:
 - $E[\phi^2(Y_t - \mu) | Y_1, \dots, Y_t] = \phi^2(Y_t - \mu)$, because $\phi^2(Y_t - \mu)$ is a function of $Y_1, \dots, Y_t$.
 - $E(\phi e_{t+1} | Y_1, \dots, Y_t) = \phi E(e_{t+1} | Y_1, \dots, Y_t) = \phi E(e_{t+1}) = 0$, because e_{t+1} is independent of $Y_1, \dots, Y_t$.
- ▶ Therefore, the *MMSE* forecast of Y_{t+2} is

$$\hat{Y}_t(2) = \mu + \phi^2(Y_t - \mu).$$

l -step ahead Forecast

- ▶ For larger lead times, the pattern continues, and the *MMSE* forecast of Y_{t+l} , for all $l \geq 1$, is given by:

$$\hat{Y}_t(l) = \mu + \phi^l(Y_t - \mu)$$

- ▶ When $-1 < \phi < 1$ (stationarity condition), note that $\phi^l \approx 0$ when l is large.
- ▶ Therefore, as l increases, the l -step ahead *MMSE* forecast:

$$\hat{Y}_t(l) \rightarrow \mu$$

- ▶ In other words, *MMSE* forecasts will "converge" to the overall process mean μ as the lead time l increases.

l -step ahead Forecast

- The *MMSE* forecast

$$\hat{Y}_t(l) \rightarrow \mu \quad \text{as } l \rightarrow \infty$$

is a characteristic of all stationary $ARMA(p, q)$ models.

Forecast Error in $AR(1)$ Model

- ▶ In the $AR(1)$ model, the 1-step ahead forecast error is

$$\begin{aligned}e_t(1) &= Y_{t+1} - \hat{Y}_t(1) \\&= (\mu + \phi(Y_t - \mu) + e_{t+1}) - (\mu + \phi(Y_t - \mu)) \\&= e_{t+1}.\end{aligned}$$

Therefore,

$$E[e_t(1)] = E(e_{t+1}) = 0, \quad \text{and} \quad \text{Var}[e_t(1)] = \text{Var}(e_{t+1}) = \sigma_e^2.$$

- ▶ Because the 1-step ahead forecast error $e_t(1)$ is an unbiased estimator of 0, we say that the 1-step ahead forecast $\hat{Y}_t(1)$ is unbiased.

Forecast Error in $AR(1)$ Model

- ▶ To find the l -step ahead forecast error, $e_t(l)$, we recall that a zero mean $AR(1)$ process can be written as an infinite order MA model, that is,

$$Y_t - \mu = e_t + \phi e_{t-1} + \phi^2 e_{t-2} + \phi^3 e_{t-3} + \cdots .$$

- ▶ Therefore,

$$Y_{t+l} - \mu = e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1} + \phi^l e_t + \cdots .$$

- ▶ The l -step ahead forecast error is

$$e_t(l) = Y_{t+l} - \hat{Y}_t(l) = Y_{t+l} - \mu + \mu - \hat{Y}_t(l).$$

Forecast Error in $AR(1)$ Model

- Replacing and expanding $Y_{t+l} - \mu$, we get:

$$e_t(l) = \underbrace{e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1} + \phi^l e_t + \cdots}_{Y_{t+l} - \mu} - \underbrace{\phi^l (Y_t - \mu)}_{\mu - \hat{Y}_t(l)}.$$

Forecast Error in $AR(1)$ Model

- Expanding $Y_t - \mu$, we obtain:

$$\begin{aligned} e_t(l) = & e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1} + \phi^l e_t + \cdots \\ & - \underbrace{\phi^l (e_t + \phi e_{t-1} + \phi^2 e_{t-2} + \phi^3 e_{t-3} + \cdots)}_{Y_t - \mu}. \end{aligned}$$

- Thus,

$$e_t(l) = e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1}.$$

Forecast Error in $AR(1)$ Model

- ▶ The l -step ahead forecast error is

$$\begin{aligned}e_t(l) &= Y_{t+l} - \hat{Y}_t(l) \\ &= e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1}.\end{aligned}$$

- ▶ Therefore, the l -step ahead forecast error has mean

$$E[e_t(l)] = E(e_{t+l} + \phi e_{t+l-1} + \phi^2 e_{t+l-2} + \cdots + \phi^{l-1} e_{t+1}) = 0,$$

i.e., forecasts are unbiased.

Forecast Error in $AR(1)$ Model

- ▶ The variance of the l -step ahead forecast error is

$$\text{Var}[e_t(l)] = \sigma_e^2 \frac{1 - \phi^{2l}}{1 - \phi^2}.$$

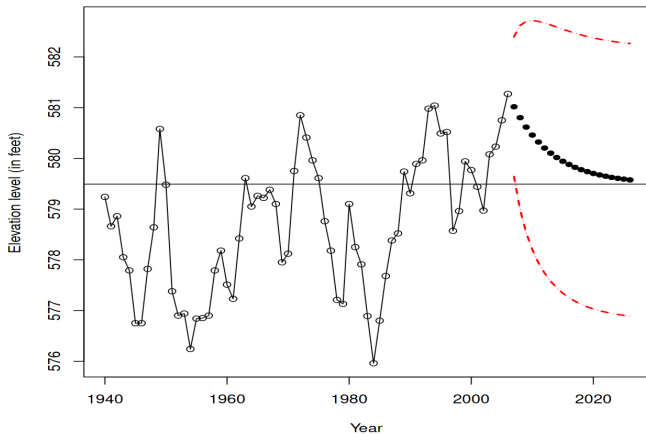
- ▶ Assuming stationarity, note that, as $l \rightarrow \infty$:

$$\text{Var}[e_t(l)] \rightarrow \frac{\sigma_e^2}{1 - \phi^2} = \gamma_0 = \text{Var}(Y_t)$$

Important

- ▶ $\text{Var}[e_t(l)] \rightarrow \gamma_0 = \text{Var}(Y_t)$ as $l \rightarrow \infty$, is a characteristic of all stationary $ARMA(p, q)$ models.

Example



Lake Huron elevation data modeled with an $AR(1)$ process and its forecast

MMSE Forecasts for $MA(1)$

- ▶ We can show, using the same approach, that the *MMSE* forecasts for the $MA(1)$ model are

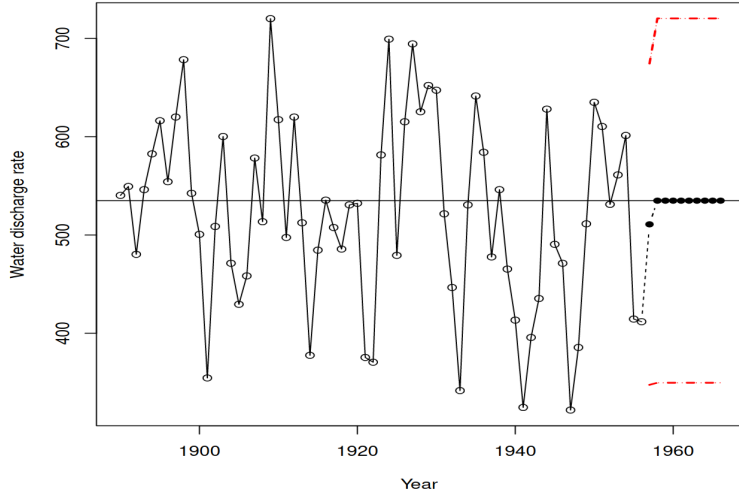
$$\hat{Y}_t(l) = \begin{cases} \mu - \theta e_t, & l = 1 \\ \mu, & l > 1 \end{cases}$$

- ▶ The key feature of an $MA(1)$ process is that observations one unit apart in time are correlated, whereas observations $l > 1$ units apart in time are not.
- ▶ For $l > 1$, there is no autocorrelation to exploit in making a prediction; this is why a constant mean prediction is made.

MMSE Forecasts for $MA(1)$

- More generally, for any purely $MA(q)$ process, the *MMSE* forecast is

$$\hat{Y}_t(l) = \mu \text{ at all lead times } l > q.$$



Gota river water discharge data modeled with an $MA(1)$ process and its forecast

ARMA(1, 1) Process

It is important to make the following observations in the *ARMA*(1, 1) case:

- ▶ The *MMSE* forecast $\hat{Y}_t(l)$ depends on the *MA* components only when $l \leq q = 1$.
- ▶ When $l > q = 1$, the *MMSE* forecast $\hat{Y}_t(l)$ depends only on the *AR* components.

This is also true of *MMSE* forecasts in higher order *ARMA*(p, q) models.

Summary of *MMSE* Forecasts for *ARMA*(p, q) Models

- ▶ When $l \leq q$, *MMSE* forecasts depend on both the *AR* and *MA* parts of the model.
- ▶ When $l > q$, the *MA* contributions vanish and forecasts will depend solely on the recursion identified in the *AR* part. That is, when $l > q$,

$$\hat{Y}_t(l) = \theta_0 + \phi_1 \hat{Y}_t(l-1) + \phi_2 \hat{Y}_t(l-2) + \cdots + \phi_p \hat{Y}_t(l-p).$$

Summary of *MMSE* Forecasts for *ARMA*(p, q) Models

It is insightful to note that the last expression, for $l > q$, can be written as

$$\hat{Y}_t(l) - \mu = \phi_1 \left[\hat{Y}_t(l-1) - \mu \right] + \cdots + \phi_p \left[\hat{Y}_t(l-p) - \mu \right].$$

Therefore,

- ▶ as a function of l , $\hat{Y}_t(l) - \mu$ follows the same Yule-Walker recursion as the autocorrelation function ρ_k .
- ▶ the roots of $\phi(x) = 1 - \phi_1 x - \phi_2 x^2 - \cdots - \phi_p x^p$ determine the behavior of $\hat{Y}_t(l) - \mu$, when $l > q$; e.g., exponential decay, damped sine waves, etc.

Summary

- ▶ For any stationary $ARMA(p, q)$ process,

$$\lim_{l \rightarrow \infty} \hat{Y}_t(l) = \mu,$$

where $\mu = E(Y_t)$. Therefore, for large lead times l , *MMSE* forecasts will be approximately equal to the process mean.

- ▶ For any stationary $ARMA(p, q)$ process, the variance of the l -step ahead forecast error satisfies

$$\lim_{l \rightarrow \infty} \text{Var}[e_t(l)] = \gamma_0,$$

where $\gamma_0 = \text{Var}(Y_t)$. That is, for large lead times l , the variance of the forecast error will be close to the process variance.

Summary

- ▶ The **predict** function in R automates the entire forecasting process, providing (estimated) *MMSE* forecasts and standard errors of the forecast error.

Nonstationary *ARIMA* Models

- ▶ For invertible *ARIMA*(p, d, q) models with $d \geq 1$, *MMSE* forecasts are computed using the same approach as in the stationary case. To see why, suppose that $d = 1$, so that the model is

$$\phi(B)(1 - B)Y_t = \theta(B)e_t,$$

- ▶ Note that

$$\phi(B)(1 - B) = \phi^*(B)$$

- ▶ We can therefore rewrite the *ARIMA*($p, 1, q$) model as

$$\phi^*(B)Y_t = \theta(B)e_t,$$

a nonstationary *ARMA*($p + 1, q$) model. We then calculate *MMSE* forecasts the same way as in the stationary case.

Example

- Suppose $p = d = q = 1$ so that we have an $ARIMA(1, 1, 1)$ process

$$(1 - \phi B)(1 - B)Y_t = (1 - \theta B)e_t.$$

- This can be written as

$$Y_t = (1 + \phi)Y_{t-1} - \phi Y_{t-2} + e_t - \theta e_{t-1},$$

a nonstationary $ARMA(2, 1)$ model.

Result

The l -step ahead forecast error $e_t(l) = Y_{t+l} - \hat{Y}_t(l)$ for any invertible $ARIMA(p, d, q)$ model has the following characteristics:

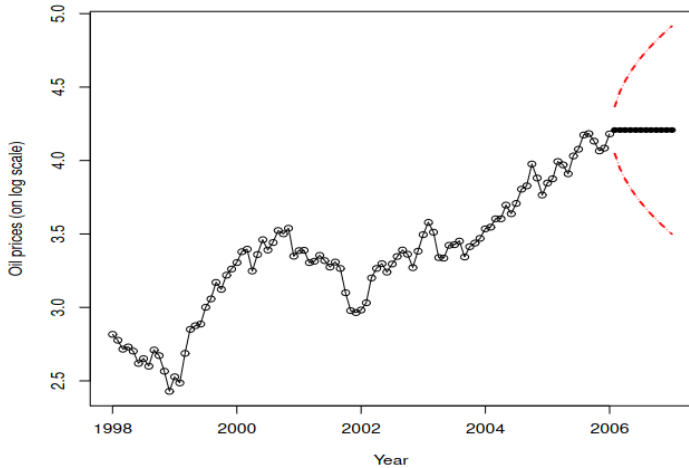
$$E[e_t(l)] = 0,$$
$$\text{Var}[e_t(l)] = \sigma_e^2 \sum_{j=0}^{l-1} \Psi_j^2,$$

where the Ψ weights correspond to those in the truncated linear process representation of the $ARIMA(p, d, q)$ model.

Result

- ▶ The first equation implies that *MMSE ARIMA* forecasts are unbiased.
- ▶ The salient feature in the second equation is that for nonstationary models, the Ψ weights do not die out as they do with stationary models.
- ▶ Therefore, for nonstationary models, the variance of the forecast error $\text{var}[e_t(l)]$ continues to increase as l does. This is not surprising given that the process is not stationary.

Example



Oil price data modeled with an $IMA(1)$ process and its forecast

ARIMA models

When working with an $ARIMA(p, d, q)$ model, two procedures are **equivalent**:

1. Forecasting the original nonstationary series Y_t .
2. Forecasting the stationary differenced series $W_t = \nabla^d Y_t$ and then summing to obtain the forecast in original terms.

ARIMA models

- ▶ The forecasts obtained from both approaches will be identical, ensuring that the associated standard errors remain the same.
- ▶ This equivalence holds because differencing is a linear operation, just as conditional expectation is.

Transformed Time Series

Recall: We previously discussed the Box-Cox family of transformations

$$T(Y_t) = \begin{cases} \frac{Y_t^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(Y_t), & \lambda = 0 \end{cases}$$

where λ is the transformation parameter.

- ▶ These transformations are not linear, thus, the forecasts are not straightforward inverse transformations of the corresponding models.
- ▶ *MMSE* forecasts are not preserved under non-linear transformations.

Transformed Time Series: log transform

- ▶ For notational purposes, set $Z_t = \ln Y_t$, and denote the *MMSE* forecast for Z_{t+l} by $\hat{Z}_t(l)$, that is, $\hat{Z}_t(l)$ is the l -step ahead *MMSE* forecast on the log scale.
- ▶ The *MMSE* forecast for Y_{t+l} is not $\hat{Y}_t(l) = \exp\{\hat{Z}_t(l)\}$!! This is sometimes called the naive forecast of Y_{t+l} .
- ▶ The theoretical argument shows that the corresponding *MMSE* forecast of Y_{t+l} is

$$\hat{Y}_t(l) = \exp \left\{ \hat{Z}_t(l) + \frac{1}{2} \text{Var}[e_t(l)] \right\},$$

where $\text{Var}[e_t(l)]$ is the variance of the l -step ahead forecast error $e_t(l) = Z_{t+l} - \hat{Z}_t(l)$.